

A Boyd-index reduction for symmetric atomic Hardy spaces

Autonomous proof attempt for arXiv:2201.10219

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Abstract

Hong–Ma–Wang ask whether their symmetric atomic Hardy space $\mathfrak{h}_{p,E}^c(\mathcal{M})$ equals the conditioned Hardy space $\mathfrak{h}_p^c(\mathcal{M})$ for $0 < p < 1$, and whether the intervening crude symmetric-atomic space also coincides. We prove that, whenever the lower Boyd index satisfies $p_E > 2$, both symmetric spaces equal the standard $(p, 2)$ -atomic space, with equivalent quasi-norms. Thus, throughout this large class of symmetric spaces, the question is exactly equivalent to the foundational unresolved problem $\mathfrak{h}_{p,\text{at}}^c = \mathfrak{h}_p^c$. The proof combines the source’s factorization identities with the constructive (p, ∞) -atomic theorem of Chen–Randrianantoanina–Xu.

1 Statement and proof intuition

Let $(\mathcal{M}, \tau)$ be a finite von Neumann algebra with normalized trace and a fixed filtration $(\mathcal{M}_n)$. Let E be a separable symmetric Banach function space on $(0, \infty)$ with $1 < p_E \leq q_E < \infty$. We use the definitions in [1]: a $(p, E)_c$ -atom a has $\mathcal{E}_n(a) = 0$, right support below some $e \in \mathcal{M}_n$, and

$$\|a\|_{\mathfrak{h}_E^c} \leq \tau(e)^{1-1/p} \|e\|_{E^\times}^{-1}. \quad (1)$$

The spaces built from these atoms and from the crude factorizations in [1, Remark 4.13] are denoted by $\mathfrak{h}_{p,E}^c$ and $\mathfrak{h}_{p,E;\text{crude}}^c$. The usual $(p, 2)$ -atomic space is $\mathfrak{h}_{p,\text{at}}^c$.

Theorem 1 (Boyd-index reduction). *Let $0 < p \leq 1$ and $2 < p_E \leq q_E < \infty$. Then*

$$\mathfrak{h}_{p,E}^c(\mathcal{M}) = \mathfrak{h}_{p,E;\text{crude}}^c(\mathcal{M}) = \mathfrak{h}_{p,\text{at}}^c(\mathcal{M})$$

with equivalent quasi-norms. The row analogues follow by adjoints, and so do the corresponding componentwise mixed-space identities.

Idea of the proof. There are two complementary factorizations. Since $p_E > 2$, the source’s Boyd-index lemma yields $L_2 = E \odot F$ for a symmetric space F ; on an e -supported operator this converts the E -bound (1) exactly into the $(p, 2)$ scaling. Conversely, choosing $q > q_E$ gives a factorization $E = G \odot L_q$, so every (p, q) atom is an E -atom. The constructive theorem of [2] decomposes every $(p, 2)$ atom into (p, ∞) atoms, closing the reverse inclusion. The same L_2 factorization and a two-norm interpolation of the adapted factor in a crude symmetric atom reduce it to a standard crude $(p, 2)$ atom.

2 The two atom comparisons

We use two standard consequences of the pointwise-product formulas quoted and proved in Section 3 of [1]. Equivalent renormings only change the constants below.

Lemma 2 (Local factorization estimates). *Assume $p_E > 2$ and write $t = \tau(e)$ for a projection e .*

1. There is a symmetric F such that $L_2 = E \odot F$ and, for $z = ze$,

$$\|z\|_2 \leq C_E \|z\|_E \frac{t^{1/2}}{\|e\|_E}. \quad (2)$$

2. For every $q > q_E$ there is a symmetric G such that $E = G \odot L_q$ and, for $z = ze$,

$$\|z\|_E \leq C_{E,q} \|z\|_q \frac{\|e\|_E}{t^{1/q}}. \quad (3)$$

Moreover $\|e\|_E \|e\|_{E^\times} = t$.

Proof. Because $p_E > 2$, Lemma 3.2 of [1] (Lemma 2.2 in its source file) renorms $E^{(1/2)}$ as a symmetric Banach space. Apply Lozanovskii factorization to $E^{(1/2)}$ and then square the product identity to obtain $L_2 = E \odot F$. The fundamental-function product formula gives $\|e\|_E \|e\|_F \asymp t^{1/2}$. Factoring $z = ze$ gives (2). For $q > q_E$, Lemma 3.3 of [1] gives $E = G \odot L_q$; now $\|e\|_{Gt^{1/q}} \asymp \|e\|_E$, and $z = ez$ gives (3). Finally $L_1 = E \odot E^\times$ gives the last identity. $\square$

If a is a $(p, E)_c$ -atom supported by e , then its conditioned square function is also supported by e [1, Lemma 4.7]. Lemma 2, (1), and the last projection identity give

$$\begin{aligned} \|a\|_{h_2^c} &\leq C_E \|a\|_{h_E^c} \frac{t^{1/2}}{\|e\|_E} \\ &\leq C_E \frac{t^{1-1/p}}{\|e\|_{E^\times}} \frac{t^{1/2}}{\|e\|_E} = C_E t^{1/2-1/p}. \end{aligned}$$

Thus every (p, E) atom is a uniformly bounded multiple of a $(p, 2)$ atom, and

$$h_{p,E}^c \hookrightarrow h_{p,\text{at}}^c. \quad (4)$$

For the converse, fix $q > q_E$. If a is a $(p, q)_c$ -atom supported by e , then (3) yields

$$\|a\|_{h_E^c} \leq C_{E,q} \|e\|_E t^{-1/q} t^{1/q-1/p} = C_{E,q} \|e\|_E t^{-1/p} = C_{E,q} \frac{t^{1-1/p}}{\|e\|_{E^\times}}.$$

So every (p, q) atom is a bounded multiple of a (p, E) atom. By [2, Theorem 4.3], every $(p, 2)$ atom is a sum of (p, ∞) atoms with uniformly controlled ℓ_p coefficients. On a support of trace t , an h_∞^c bound implies the required h_q^c bound by Hölder, so these are also (p, q) atoms. Consequently

$$h_{p,\text{at}}^c \hookrightarrow h_{p,E}^c. \quad (5)$$

Equations (4)–(5) prove the first and third spaces in Theorem 1 are equal.

3 The crude symmetric atoms

It remains to control the middle space. For $p = 1$, [1] records $h_{1,E}^c = h_1^c$ and

$$h_{1,E}^c \subset h_{1,E;\text{crude}}^c \subset h_1^c,$$

so equality is immediate. Assume $0 < p < 1$, and put

$$u = \frac{p}{1-p}, \quad s = \frac{2p}{2-p}, \quad \frac{1}{p} = \frac{1}{2} + \frac{1}{s}.$$

Let $a = yb$ be a crude $(p, E)_c$ -atom. Thus $\mathcal{E}_n(y) = 0$, $\|y\|_{\mathfrak{h}_E^c} \leq 1$, and

$$b \in L_u(\mathcal{M}_n) \cap E^\times(\mathcal{M}_n), \quad \|b\|_u \|b\|_{E^\times} \leq 1. \quad (6)$$

Taking $e = 1$ in (2) gives $\|y\|_{\mathfrak{h}_2^c} \leq C_E$. Choose $q > q_E$. The continuous inclusion $L_q \hookrightarrow E$ dualizes to $E^\times \hookrightarrow L_{q'}$, hence $\|b\|_{q'} \leq C_{E,q} \|b\|_{E^\times}$. Define v by

$$\frac{2}{v} = \frac{1}{u} + \frac{1}{q'} = \frac{1}{p} - \frac{1}{q}.$$

Noncommutative Hölder applied to $|b|^2$ gives

$$\|b\|_v^2 = \| |b|^2 \|_{v/2} \leq \|b\|_u \|b\|_{q'} \leq C_{E,q}. \quad (7)$$

But

$$\frac{1}{s} - \frac{1}{v} = \frac{1-p}{2p} + \frac{1}{2q} > 0,$$

so $v > s$. Since $\tau(1) = 1$, $L_v \hookrightarrow L_s$ contractively. Equations (6)–(7) show that a is a uniformly bounded multiple of a standard crude $(p, 2)_c$ -atom. Proposition 3.9 of [2] decomposes every such crude atom into $(p, 2)$ atoms with controlled ℓ_p coefficients. Therefore

$$\mathfrak{h}_{p,E;\text{crude}}^c \hookrightarrow \mathfrak{h}_{p,\text{at}}^c.$$

Together with the defining inclusion $\mathfrak{h}_{p,E}^c \subset \mathfrak{h}_{p,E;\text{crude}}^c$ and (5), this completes the proof of Theorem 1.

4 What remains open

For $0 < p < 1$, [2] proves $\mathfrak{h}_{p,\text{alg}}^c = \mathfrak{h}_p^c$ but explicitly leaves open $\mathfrak{h}_{p,\text{at}}^c = \mathfrak{h}_p^c$. The weak decomposition there controls coefficients only in ℓ_1 , whereas the atomic quasi-norm requires ℓ_p control. Theorem 1 therefore gives, for every E with $p_E > 2$,

$$\mathfrak{h}_{p,E}^c = \mathfrak{h}_p^c \iff \mathfrak{h}_{p,\text{at}}^c = \mathfrak{h}_p^c.$$

This is a substantial reduction but not a solution of the foundational right-hand equality. At the endpoint $p_E = 2$, the proof still applies whenever the structural factorization $L_2 = E \odot F$ with matching fundamental functions is available (in particular for $E = L_2$); the Boyd index alone no longer supplies it.

References

- [1] G. Hong, C. Ma, and Y. Wang, *John–Nirenberg inequalities for noncommutative column BMO and Lipschitz martingales*, arXiv:2201.10219v1 (2022).
- [2] Z. Chen, N. Randrianantoanina, and Q. Xu, *Atomic decompositions for noncommutative martingales*, arXiv:2001.08775v1 (2020).